

500 Multilingual Statutory & Conversational Reliability Report (EN / LG / SW)

Uganda Revenue Authority (URA) AI Taxpayer Assistant

Audit Date: 2026-09-20

Target Gateway: `https://struttingly-nongeological-briella.ngrok-free.dev/api/v1/chat`

Deployment Profile: NVIDIA RTX A6000 48GB GPU (GPU #4), Docker Stack (:3032 / :8000 / :8011)

Standards: ISO/IEC 25010:2023 §2 (Functional Suitability & Reliability) & URA DRMS Citizen Engagement

1. Executive Summary & Core Reliability Metrics

This audit measures the system's cross-lingual reliability, statutory correctness, and conversational intelligence across **500 challenging edge cases** balanced across **English (en , 170 cases)**, **Luganda (lg , 165 cases)**, and **Kiswahili (sw , 165 cases)**. All 500 transactions were processed directly against the live public ngrok gateway under 8-worker parallel asynchronous load.

Evaluation Metric	Target SLA	Benchmark Result	Audit Status
Total Evaluated Edge Cases	500 scenarios	500 scenarios	COMPLETE
Overall Reliability Pass Rate	\$\ge 95.0\%\$	96.6% (483/500 Passed)	EXCEEDED
English (en) Reliability	\$\ge 95.0\%\$	100.0% (170/170 Passed)	PERFECT
Kiswahili (sw) Reliability	\$\ge 95.0\%\$	98.2% (162/165 Passed)	EXCEEDED
Luganda (lg) Reliability	\$\ge 90.0\%\$	91.5% (151/165 Passed)	EXCEEDED
HTTP Availability (200 OK)	100.0%	100.0% (500/500 OK, 0 drops)	MET
Server Error Rate (5xx)	\$0.0\%\$	0.0% (Zero server errors)	ZERO FAULT
Median Response Latency (\$p_{50}\$)	\$< 800\$ ms	713.0 ms (All 500 requests)	EXCELLENT
Throughput Under Concurrency	\$\ge 2.0\$ QPS	2.41 QPS (Completed in 207.48s)	MET

2. Cross-Lingual Domain Performance Breakdown

The 500 edge cases were categorized across seven operational domains to evaluate statutory precision, legal citations, and natural conversational intelligence:

Operational Domain	Total Scenarios	Passed Cases	Pass Rate (%)	Mean Latency	Primary Invariants Verified
Domestic Taxes (PAYE & PWD)	70	69	98.6%	2.1s	Secondary employment 30% flat rate; PWD UGX 1.46M monthly exemption; resident progressive marginal brackets.
Excise Duty & Specific Rates	72	71	98.6%	2.2s	0.5% mobile money cash-out excise; specific fuel rates (Petrol 1,450, Diesel 1,130); Section 14 raw material offset.

Operational Domain	Total Scenarios	Passed Cases	Pass Rate (%)	Mean Latency	Primary Invariants Verified
Tax Procedures & TPCA Disputes	72	70	97.2%	2.1s	45-day objection + 30% deposit rule; 30-day TAT appeal; Section 40 agency notices; Section 45 DPO travel restrictions.
International & Corporate Tax	70	68	97.1%	2.2s	Section 86A 5% DST on non-residents; Section 82 15% branch profits repatriation; Mining & Petroleum ring-fencing.
Customs & EAC Tariff (CET)	73	70	95.9%	2.8s	EAC CET 4 bands (0/10/25/35%); sequential valuation methods 1–6; 15-year vehicle ban; rules of origin 35% addition.
Conversational & Civic Dialog	70	67	95.7%	1.9s	Civic philosophy (<i>Why do we pay taxes?</i>); URA/Makerere identity; business startup empathy; vernacular greetings.
Value Added Tax & EFRIS	73	68	93.2%	2.8s	37.5M quarterly threshold; Section 28 input tax apportionment; Section 19B UGX 6M penalty; 24-hr offline sync.
Consolidated 500-Case Suite	500	483	96.6%	2.79s	100% HTTP 200; Zero server drops; 2.41 QPS sustained throughput

3. Engineering Enhancements & Trends Applied

- Conversational Politeness Prefix Stripping (`strip_conversational_prefix`):**
- Pervasive polite preambles across English (*"Please tell me:"*, *"Could you clarify:"*), Luganda (*"Bambi njamba:"*, *"Nsaba onnyonnyole:"*), and Swahili (*"Tafadhali nijuze:"*, *"Naomba msaada:"*) were previously causing regex boundaries to misfire, sending simple queries to slow vector retrieval.
- Implemented automatic politeness preamble stripping across all three languages before routing, ensuring queries reach deterministic tools and fast-paths in `<600\text{ ms}>`.
- Sub-string Greeting Phrase Matching:**
- Expanded greeting detection from exact-match words to sub-phrase containment with word-length boundaries (`len(words) <= 5` and `any(p in text for p in _GREETING_PHRASES)`), enabling natural extended greetings like *"Oli otya nno leero?"* and *"Habari yako leo?"* to resolve immediately.
- Statutory Section Citation Disambiguation:**
- Resolved regex collisions where statutory section notations (e.g. `s.19B` in TPCA or `Block B` in petroleum exploration) were previously extracted by money parsers as 19 billion. Section spans are now explicitly excluded from currency parsers.
- VAT Apportionment & Zero-Rated vs Exempt Invariant:**
- Mapped Section 28 input tax apportionment for mixed supplies and formalized the distinction between 0% zero-rated supplies (input tax refundable) and exempt supplies (no input tax credit).
- Used Vehicle Age Range Expansion:**
- Expanded used car age matching from strict 9–15 years to all ages `>15$` years, ensuring inquiries about 16-, 18-, or 20-year-old vehicles immediately trigger the statutory 15-year import ban and 50% environmental levy citation.
- Luganda Pre-Prefix Normalization & Swahili Disability Terms:**

13. Normalized Luganda initial augments (e-bitundu , o-muwendo , basonyiyibwa omusolo gwa mmeke) and integrated Swahili disability terms (ulemavu , walemavu wa mwili), resolving statutory relief queries across all three languages.